

Rough Total

\$102,554.68

[illegible]

1/4" ID PTFE Tubing	VWR/Thermo Scientif	63014-713	scrubber	\$4.81	25 ft
HF Fumes Detector	VWR/GfG Instrument	76334-718	electronic	\$1,468.45	2
HF Gas Cylinder	Linde	UN1052	gas	\$2,702.84	1
Ar Gas Cylinder	Brazos Welding		gas		
Scale	Mettler Toledo	H-5275	electronic	\$895.00	
Thermocouple	Pyromation	GP02	electronic	\$285.10	1
3/4" to 1/2" 316SS Union	Swagelok	SS-1210-6-8	pipng	\$41.10	6
1/2" to 3/8" 316SS Union	Swagelok	SS-810-6-6	pipng	\$26.90	9
1/2" to 1/4" 316SS Union	Swagelok	SS-810-6-4	pipng	\$27.00	9
3/8" to 1/4" 316SS Union	Swagelok	SS-600-6-4	pipng	\$17.40	5
1/2" C-276 Union	Swagelok	HC-810-6	pipng	\$526.90	1
1/4" 316SS Cross Union	Swagelok	SS-400-4	pipng	\$45.30	6
1/2" OD x 25" L x 1/8" T Ni-200 Tube	Magellan Metals		pipng	\$1,605.00	1
3/8" OD x 23" L x 1/16" T Ni-200 Tube	Magellan Metals		pipng	\$797.00	2
1/8" OD x 4" L x 1/32" T Ni-200 Tube	Magellan Metals		pipng	\$775.00	1
28" L x 18" W x 1/8" T Ni-200 Plate	Magellan Metals		vessel	\$830.00	1
8" FOD x 16" L 316SS Half Nipple	Kurt Lesker		vessel	\$2,248.55	1
316SS Plate	McMaster Carr		vessel	\$114.70	1
Residual Gas Analyzer	Stanford Research Systems		electronic	\$3,890.65	1
Fire Brick	Lynn Manufacturing	K-23	heating	\$9.33	12
Ceramic Fiber Blanket	Lynn Manufacturing		heating	\$24.90	1
C-276 Rupure Disk Housing	Alpha Process Sales		vessel	\$1,896.00	3
^ machining costs	BS&B Safety Systems		vessel	\$1,300.00	1
Thermocouple	Parr Instruments	A472E5	electronic	\$80.00	1
Heating Tape	Brisk Heat	TBIH051-040-LSE	heating	\$125.00	1
PTFE Bottle	Lab Depot	314874	scrubber	\$245.61	1

Hydrogen Generator	Peak Scientific	3301768	gas	\$23,623.20	1
1/4" 316SS Needle Valve	Swagelok	SS-1RS4	pipng	\$80.01	1
1/4" 316SS Check Valve	Swagelok	SS-4C-KZ-1	pipng	\$98.91	3
Turbo Pump	Leybold	Turbovac 90i / DN 63 Cf	vacuum	\$4,169.00	1
Turbo Pump Power Supply and Cables	Leybold	Turbo.power Integra	vacuum	\$777.15	1
1/2" 316SS Needle Valve	Swagelok	SS-6NBS8(-G)	pipng	\$253.00	5
0.3 mm ID PTFE Tubing	Omnifit		pipng	\$64.81	1
1/4" Kynar Tubing	Ultra Pure Products	RFPVDF-.187-250	pipng	\$1.41	20 ft
Oxygen Trap	VWR/Restek	20638	gas	\$372.80	2
1/4" to 1/16" 316SS Union	Swagelok	SS-400-6-1FGC	pipng	\$38.22	2
4.5" CF 304SS Blank Flange	Kurt Lesker	F0450X000N	vacuum	\$59.05	1
KF16 Flexible Metal Hose	LDS Vacuum	NW16-MH48-VE	vacuum	\$214.00	1
KF16 to 1/4" Swagelok Adapter	LDS Vacuum	NW16x25SWG-VE	vacuum	\$57.00	2
KF25 to KF16 Union	LDS Vacuum	NW25x16-CR-VE	vacuum	\$49.00	1
KF25 to 1/4" Swagelok Adapter	LDS Vacuum	NW25x25SWG-VE	vacuum	\$69.00	1
2.75" CF 304SS Blank Flange	LDS Vacuum	275-000-VE	vacuum	\$21.00	1
2.75" CF to 1/4" Swagelok Adapter	LDS Vacuum	275xSWG-25	vacuum	\$69.00	1
4.5" CF to 2.75" CF Reducer	LDS Vacuum	450x275-150-0-T1	vacuum	\$104.00	1
4.5" CF to 2.75" CF Reducing Cross	LDS Vacuum	450x275-RDX	vacuum	\$300.00	1
KF16 to 1/2" Swagelok Adapter	LDS Vacuum	NW16x50SWG-VE	vacuum	\$73.00	2
KF25 to KF16 Flexible Metal hose	LDS Vacuum	NW25x16-MH20-VE	vacuum	\$162.00	1
Roughing Pump	A-VAC industries	Edwards RV5	vacuum	\$1,630.00	1
1/4" Compression to 3/8" Male NPT PTFE Union	VWR/Ace Glass	12710-12	scrubber	\$52.50	2
1/4" 316SS Check Valve	Swagelok	SS-CHS4-1	pipng	\$61.40	3
1/4" Swagelok to 1/4" Male NPT 316SS Union	Swagelok	SS-400-1-4	pipng	\$7.90	10
1/4" Swagelok 316SS Elbow Union	Swagelok	SS-400-9	pipng	\$17.70	10
1/4" Swagelok 316SS Tee Union	Swagelok	SS-400-3	pipng	\$24.90	10
1/4" 316SS Ball Valve	Swagelok	SS-42GS4	pipng	\$98.76	2
1/4" 316SS Ball Valve	Swagelok	SS-43GS4	pipng	\$99.40	5
1/4" 316SS 3-way Ball Valve	Swagelok	SS-43GXS4	pipng	\$125.00	5
1/2" 316SS Ball Valve	Swagelok	SS-45S8	pipng	\$234.10	3

1/2" C-276 Needle Valve	Swagelok	HC-6NBS8-G-HC	pipng	\$2,024.26	2
1/4" 316SS Tubing	Swagelok	SS-T4-S-035-20	pipng	\$3.12	60 ft
3/4" Swagelok 316SS Union	Swagelok	SS-1210-6	pipng	\$42.20	5
C-276 Plate	Magellan Metals		vessel	\$315.00	2
1/4" CS Tube	Earle M. Jorgensen Co		pipng	\$6.25	40 ft
Corrosive Gas Regulator	Matheson		gas	\$2,201.24	1
KF16 Ball Valve	Kurt Lesker	BVB0075MVQF	vacuum	\$345.00	1
Ion Pressure Gauge	Kurt Lesker	KJLC354401YF	electronic	\$1,308.00	1
I/O Controller	National Instruments	781441-01	electronic	\$2,745.90	1
1/4" to 1/8" 316SS Union	Swagelok	SS-400-6-2	pipng	\$14.18	1
Heating Tape w/ Controller	Brisk Heat	HSTAT051004	heating	\$199.56	1
Ceramic Fiber Insulation	Unitherm	CF8-5-24x60in	heating	\$40.53	1
1/4" Female NPT to Swagelok Monel adapter	Swagelok	M-400-7-4	pipng	\$44.35	1

Extra Comments

for H2: THERMAL MFC 316 SS PTFE/KALREZ 1/4"COMPR
FTG LCD DISPLAY 0-5VDC/4-20MA Hydrogen H2 0-2 STD
L/MIN STD METERING PRESS [14.7PSIA] STD TEMP [21.1
°C/70°F]

for Ar: THERMAL MFC ALUMINUM BUNA-N 1/4"COMPR
FTG LCD DISPLAY 0-5VDC/4-20MA Argon Ar 0-2 STD
L/MIN STD METERING PRESS [14.7PSIA] STD TEMP [21.1
°C/70°F]

Standard output, fittings, display, working pressure, &
calibration type

316SS mesh; for 1/2" tube

Tube type construction with stainless steel shell.

-Dimensions: 8.25 in. I.D. x 16 in. O.D. x 13 in. Long

-Elements: Kanthal A1, embedded windings.

-Furnace Temperature Rating: 800 deg. C.

-Single Zone Construction, 10 in. heated length

-Total Power: 5350 Watts/23.2 amps @ 230 VAC, 1Ph, 60
Hz.

-End Disk: 1.5 in. thick, bore 7 in. both ends

-End Caps: 1.5 in. thick, bore 7 in. both ends

-(1) Thermocouple Port: Located midlength of heated
zone

-(1) Thermocouple, Type K, with ceramic insulator

-(1) Thermocouple, Type K, connector with mounting
bracket

-(1 set) Support Brackets for vertical mount

Possession Status**Used in Build****Lead Time****Company Hassle**

arrived

yes

long

no

arrived

yes

long

no

arrived

yes

very long

custom part

arrived

yes

very long

no

arrived

no

very long

no

arrived

yes

very long

custom part

arrived

yes

long

no

arrived

yes

long

no

arrived

no

medium

yes

arrived

yes

medium

yes

Range: 0-10 ppm, Resolution: 0.1 ppm
 4.5 L (3.8kg) filled, 30" tall with valve, 4.5" diameter

Mettler-Toledo™ BC60 Parcel Scale - 150 lbs x .05 lb
 K-type, 16in long sheath, 1/4in diameter sheath, 1/2in
 screw connection, 4-20 mA output, 0-900C range

8"OAL, 10"CF FIXED FLANGE, 8"OD X .120" WALL
 TUBING. OPEN END OF TUBE WILL BE UNCAPPED.
 FLANGE SIDE OF TUBING WILL BE BLANKED
 MATERIAL: 316LN SS FLANGE WITH TUBING/DISC FROM
 316/316L SS
 FINISH: POLISHED TUBE. CAPPED END OF TUBE WILL BE
 EITHER DA FINISH OR BRUSH FINISH.

8" L x 8" W x 1/4" T

RGA 100

9" x 4.5" x 2.5"

25" x 24" x 0.5"

1/2" Swagelok Connection, 80 psi, 50C, 316SS outlet

Thermocouple, type-J, 21-1/2" 1/8, T316

120 VAC, max temp 900°F, leads same ends, 2 ft

500 mL

arrived	no	medium	yes
arrived	yes	long	yes
arrived	yes	long - very long	supply issues
arrived	yes	short	no
arrived	yes	medium	no
arrived	no	medium	no
arrived		1 short	no
arrived		2 short	no
arrived		2 short	no
arrived		1 short	no
arrived	yes	long	no
arrived		1 short	no
arrived	yes	long	no
arrived	yes	long	no
arrived	yes	long	no
arrived	yes	medium	no
arrived	yes	long	yes & custom part
arrived	yes	short	no
arrived	yes	medium	no
arrived	yes	short	no
arrived	yes	short	no
arrived	yes	very long	yes & custom part
		very long	yes & custom part
arrived	yes	short	no
arrived	yes	medium	yes
arrived	yes	medium	no

Trace 1200cc, 100-230V, 8L bottle	arrived	yes	very long	no
	arrived	yes	short	no
Kynar Seals, 1 psi	arrived	yes	medium	no
4.5" CF flange inlet, KF16 outlet	arrived	yes	long	no
	arrived	yes	long	no
2 have grafoil packing	arrived	yes	short	no
20m length, 1/16" OD	arrived	yes	medium	no
	arrived	yes	medium	no
1/4" swagelok connections	arrived	yes	medium	yes
	arrived	yes	short	no
	arrived	yes	medium	no
48" length, Sealing Rings and clamps included	arrived	yes	medium	no +
Sealing Rings and clamps included	arrived	yes	medium	no +
Sealing Rings and clamps included	arrived	yes	medium	no +
Sealing Rings and clamps included	arrived	yes	medium	no +
Nuts, bolts, and washers included	arrived	yes	medium	no +
Nuts, bolts, and washers included	arrived	yes	medium	no +
Nuts, bolts, and washers included	arrived	yes	medium	no +
Nuts, bolts, and washers included	arrived	yes	medium	no +
Sealing Rings and clamps included	arrived	yes	medium	no +
20" length, Sealing Rings and clamps included	arrived	no	medium	no +
Rebuilt, 4.1 cfm, 1ph, 115/230V	arrived	yes	medium	no +
	arrived	yes	long	no
1 psi	arrived	no	short	no
	arrived	no	short	no
	arrived	yes	short	no
	arrived	yes	short	no
1 piece, 0.6 Cv	arrived	yes		
1 piece, 1.4 Cv	arrived	yes (get more)	short	no
1 piece, 0.9 Cv	arrived	no	short	no
1 piece, 12 Cv	arrived		2 short	no

Grafoil packing	arrived	yes	short	no
in 6ft sections, 0.035" wall thickness, 5100 psi	arrived	yes	short	no
	arrived	no	short	no
0.25" x 2.5" x 6"	arrived	yes	medium	no
	arrived	no	medium	yes
Monel, Single Stage, 1-50 psi, 660 CGA	arrived	yes	long	yes
	arrived	yes	long	no
2.75" CF connection	arrived	yes	medium	no
NI USB-6353, X Series DAQ (32 AI, 48 DIO, 4 AO) & Power C	ordered	yes	very long	Custom part
for connecting up the hydrogen generator	arrived	yes	short	no
4 ft long, 0.5 in wide, 120 V, 144 Watt, 10-218° C	arrived	yes	short	no
5 ft long, 2 ft wide, 0.5 in thick	arrived	no	short	no
	ordered	yes	short	no

short	1-2 days
medium	weeks
long	month

very long 2 months+

